

Week 4, Lecture 2 WOD – Answer Key

Expectation and the Sample Mean

NRES 746 – Instructor Answer Key (not for student distribution)

This WOD is ungraded, so this key isn't a grading rubric – it's a quick reference for circulating during the 8-10 minute window. Questions with a single correct derivation or number are answered directly; the one genuinely open-ended item (2b) is given a sample answer and discussion points rather than an exact-match rubric.

Part 1: A parameter vs. a statistic

1. μ is **not** a random variable – it's a fixed (if unknown) property of the entire population of trout, and it does not change no matter who does the sampling. $\bar{x}$ is a random variable – it's computed from whichever 5 trout happen to get caught, so a different sample of 5 would give a different $\bar{x}$. This is the core distinction between a **parameter** (a fixed number describing a population, e.g. μ , σ^2) and a **statistic** (a number computed from a sample, e.g. $\bar{x}$, which is itself a random variable with its own distribution – exactly what Parts 3-4 investigate).

Part 2: Working with expectations

2a.

$$\mathbb{E}[2X + 3] = 2\mathbb{E}[X] + 3 = 2(22) + 3 = 47 \quad \text{Var}(2X + 3) = 2^2\text{Var}(X) = 4(16) = 64$$

```
mu <- 22; sigma2 <- 16
2 * mu + 3          # E[2X+3]

## [1] 47

2^2 * sigma2        # Var(2X+3)

## [1] 64
```

Common mistakes to watch for: (1) forgetting to square the coefficient on the variance (writing $\text{Var}(2X + 3) = 2(16) = 32$ instead of $2^2(16) = 64$); (2) trying to fold the $+3$ into the variance calculation at all (e.g. $2(16) + 3$) – the additive constant shifts the mean but has no effect on spread.

2b. The special condition is **independence** of X and Y . Students aren't expected to derive this algebraically yet (that requires covariance, which the course hasn't covered), but "independent" is a reasonable guess since the problem explicitly describes X and Y as independent measurements, and Part 3 relies on this same property.

Instructor note, for context: the *exact* necessary condition is a bit weaker than full independence – $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ holds whenever X and Y are merely **uncorrelated** ($\text{Cov}(X, Y) = 0$), which independence guarantees but doesn't require. Worth mentioning if a student pushes on this, but not worth deriving cold in an 8-10 minute WOD – flag it as a preview for whenever covariance gets formally introduced.

Part 3: Averaging two measurements

3a. $M = \frac{1}{2}X_1 + \frac{1}{2}X_2$, so $a = b = \frac{1}{2}$.

$$\mathbb{E}[M] = \frac{1}{2}\mathbb{E}[X_1] + \frac{1}{2}\mathbb{E}[X_2] = \frac{1}{2}\mu + \frac{1}{2}\mu = \mu$$

For the variance, apply $\text{Var}(aX) = a^2\text{Var}(X)$ to each term, then add (valid because X_1, X_2 are independent, per Part 2b):

$$\text{Var}(M) = \text{Var}\left(\frac{1}{2}X_1\right) + \text{Var}\left(\frac{1}{2}X_2\right) = \frac{1}{4}\text{Var}(X_1) + \frac{1}{4}\text{Var}(X_2) = \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2 = \frac{\sigma^2}{2}$$

Common mistake to watch for: forgetting to square the $\frac{1}{2}$ coefficients, which gives $\text{Var}(M) = \frac{1}{2}\sigma^2 + \frac{1}{2}\sigma^2 = \sigma^2$ (no reduction at all) instead of the correct $\sigma^2/2$ – this is the exact error Part 3a’s “name a and b ” scaffolding is designed to catch.

3b.

```
sigma2 / 2
```

```
## [1] 8
```

$\text{Var}(M) = 8$, compared to $\text{Var}(X_1) = 16$ alone – **half**. Averaging two independent measurements cuts the variance in half (increases precision) relative to a single measurement.

Part 4: Generalizing to n measurements

4a. $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n}X_1 + \frac{1}{n}X_2 + \cdots + \frac{1}{n}X_n$, i.e. each term has coefficient $\frac{1}{n}$.

$$\mathbb{E}[\bar{X}] = \sum_{i=1}^n \frac{1}{n} \mathbb{E}[X_i] = \sum_{i=1}^n \frac{1}{n} \mu = n \cdot \frac{\mu}{n} = \mu$$

$$\text{Var}(\bar{X}) = \sum_{i=1}^n \text{Var}\left(\frac{1}{n}X_i\right) = \sum_{i=1}^n \frac{1}{n^2} \text{Var}(X_i) = n \cdot \frac{\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

(the sum-of-variances step again requires independence across all n measurements.)

4b.

```
sigma2 / 5
```

```
## [1] 3.2
```

$\text{Var}(\bar{X}) = 3.2$ at $n = 5$. Checking against Part 3b: plugging $n = 2$ into σ^2/n gives $16/2 = 8$, matching Part 3b’s answer exactly – a good internal consistency check that the general formula is correct.

4c. As $n \rightarrow \infty$, $\text{Var}(\bar{X}) = \sigma^2/n \rightarrow 0$. In plain language: with a very large sample, the sample mean becomes an arbitrarily precise estimate of the true population mean μ – the more data you collect, the less your particular sample’s average is likely to stray from the true value.

4d.

$$\text{SD}(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \sqrt{\sigma^2/n} = \frac{\sigma}{\sqrt{n}}$$

```
sqrt(sigma2) / sqrt(5)
```

```
## [1] 1.788854
```

The standard deviation of $\bar{X}$ (the standard error) is about **1.79** at $n = 5$, and it scales as $1/\sqrt{n}$, not $1/n$ – a genuinely different rate than the variance. **Common mistake to watch for:** students conflating these two formulas (e.g. writing $\text{SD}(\bar{X}) = \sigma^2/n$, or claiming the standard error halves whenever n doubles). Worth spelling out explicitly: to cut the standard error in half, you need to **quadruple** n , not double it – precision improves more slowly than students’ intuition from the variance formula alone would suggest. This is the standard “law of diminishing returns” behind why doubling a study’s sample size rarely doubles its precision.